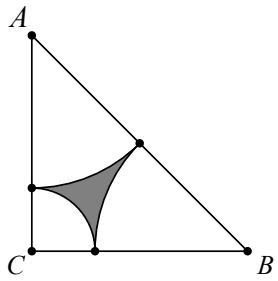
2012 Team Problems

T-1. In $\triangle ABC$, $\text{m}\angle A = \text{m}\angle B = 45^\circ$ and $AB = 16$. Mutually tangent circular arcs are drawn centered at all three vertices; the arcs centered at A and B intersect at the midpoint of $\overline{AB}$. Compute the area of the region inside the triangle and outside of the three arcs.



T-2. Compute the number of ordered pairs of integers (a, b) such that $1 < a \leq 50$, $1 < b \leq 50$, and $\log_b a$ is rational.

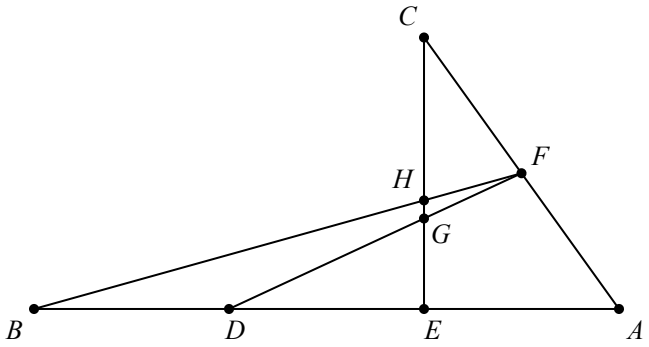
T-3. Suppose that 5-letter "words" are formed using only the letters A, R, M, and L. Each letter need not be used in a word, but each word must contain at least two distinct letters. Compute the number of such words that use the letter A more than any other letter.

T-4. Positive integers $a_1, a_2, a_3, \ldots$ form an arithmetic sequence. If $a_1 = 10$ and $a_{a_2} = 100$, compute $a_{a_{a_3}}$.

T-5. The graphs of $y = x^2 - |x| - 12$ and $y = |x| - k$ intersect at distinct points A, B, C, and D, in order of increasing x-coordinates. If $AB = BC = CD$, compute k.

T-6. The zeros of $f(x) = x^6 + 2x^5 + 3x^4 + 5x^3 + 8x^2 + 13x + 21$ are distinct complex numbers. Compute the average value of $A+BC+DEF$ over all possible permutations (A, B, C, D, E, F) of these six numbers.

T-7. Given noncollinear points A, B, C, segment $\overline{AB}$ is trisected by points D and E, and F is the midpoint of segment $\overline{AC}$. $\overline{DF}$ and $\overline{BF}$ intersect $\overline{CE}$ at G and H, respectively. If $[DEG] = 18$, compute $[FGH]$.



T-8. Let $N = \left\lfloor (3+\sqrt{5})^{34} \right\rfloor$. Compute the remainder when N is divided by 100.

T-9. Let ABC be a triangle with $\mathrm{m}\angle B = \mathrm{m}\angle C = 80^\circ$. Compute the number of points P in the plane such that triangles PAB, PBC, and PCA are all isosceles and non-degenerate. Note: the approximation $\cos 80^\circ \approx 0.17$ may be useful.

T-10. If $\lceil u \rceil$ denotes the least integer greater than or equal to u, and $\lfloor u \rfloor$ denotes the greatest integer less than or equal to u, compute the largest solution x to the equation

$$\left\lfloor \frac{x}{3} \right\rfloor + \lceil 3x \rceil = \sqrt{11} \cdot x.$$